

Procedures

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1 Introduction

This document explains how to do some of the exercises that you have done during the course. The idea is to help you get better at maths by practising the steps. I hope this helps you. This document is going to keep changing, so please be patient.

2 Property of logarithms

Laws of Logarithms

$$\log_a (AB) = \log_a (A) + \log_a (B)$$

$$\log_a \left(\frac{A}{B} \right) = \log_a (A) - \log_a (B)$$

$$\log_a (A^C) = C \log_a (A)$$

Problem 1: Reducing logarithmic expressions

$$\begin{aligned} & \frac{1}{6} [\log_8 [x] + \log_8 [y+4]] - 2 \log_8 [z-1] \\ & \frac{1}{6} [\log_8 [x] + \log_8 [y+4]] - \log_8 [(z-1)^2] \\ & \frac{1}{6} \log_8 [x] + \frac{1}{6} \log_8 [y+4] - \log_8 [(z-1)^2] \\ & \log_8 [x^{1/6}] + \log_8 [(y+4)^{1/6}] - \log_8 [(z-1)^2] \\ & \log_8 \left[(x^{1/6}) ((y+4)^{1/6}) \right] - \log_8 [(z-1)^2] \\ & \log_8 [(x(y+4))^{1/6}] - \log_8 [(z-1)^2] \\ & \log_8 \left[\frac{(x(y+4))^{1/6}}{(z-1)^2} \right] \end{aligned}$$

$$\frac{1}{6} [\log_8 [x] + \log_8 [y + 4]] - 2 \log_8 [z - 1] = \log_8 \left[\frac{(x(y + 4))^{1/6}}{(z - 1)^2} \right]$$

3 Solving equations

In this section are some solutions of different types of equations.

Problem 2: Example of exponential equation

We are going to solve the following equation

$$8^{x-2} = 3^{x+3}.$$

Since is an exponential function we apply a logarithmic function in order to apply logarithmic properties to manipulate the expression.

$$\begin{aligned} \ln [8^{x-2}] &= \ln [3^{x+3}] \\ (x-2) \ln [8] &= (x+3) \ln [3] \\ x \ln [8] - 2 \ln [8] &= x \ln [3] + 3 \ln [3] \\ x \ln [8] - \ln [8^2] &= x \ln [3] + \ln [3^3] \\ x \ln [8] - \ln [8^2] + \ln [8^2] &= x \ln [3] + \ln [3^3] + \ln [8^2] \\ x \ln [8] &= x \ln [3] + \ln [3^3] + \ln [8^2] \\ x \ln [8] - x \ln [3] &= x \ln [3] - x \ln [3] + \ln [3^3] + \ln [8^2] \\ x \ln [8] - x \ln [3] &= \ln [3^3] + \ln [8^2] \\ x (\ln [8] - \ln [3]) &= \ln [3^3] + \ln [8^2] \\ x \ln \left[\frac{8}{3} \right] &= \ln [3^3 \cdot 8^2] \\ x \ln \left[\frac{8}{3} \right] \frac{1}{\ln \left[\frac{8}{3} \right]} &= \ln [3^3 \cdot 8^2] \frac{1}{\ln \left[\frac{8}{3} \right]} \\ x &= \frac{\ln [3^3 \cdot 8^2]}{\ln \left[\frac{8}{3} \right]} \approx 7.600425789 \end{aligned}$$

$$\text{Answer : } x = \frac{\ln [1728]}{\ln \left[\frac{8}{3} \right]} \approx 7.600425789$$

Problem 3: Example of logarithmic equation

We are going to solve the following equation

$$\log_2 [x + 1] + \log_2 [x - 1] = 3.$$

Let's start by reducing the logarithms and then solve for x ,

$$\begin{aligned}\log_2 [x + 1] + \log_2 [x - 1] &= 3 \\ \log_2 [(x + 1)(x - 1)] &= 3 \\ 2^{\log_2 [(x + 1)(x - 1)]} &= 2^3 \\ (x + 1)(x - 1) &= 2^3 \\ x^2 - x + x - 1 &= 2^3 \\ x^2 - 1 &= 2^3 \\ x^2 - 1 + 1 &= 2^3 + 1 \\ x^2 &= 9 \\ \sqrt{x^2} &= \pm\sqrt{9} \\ x &= \pm 3\end{aligned}$$

Answer : $x = \pm 3$

3.1 Critical points

To find the critical point we need to remember that for exponential functions we equate the argument to 0 and then for x . On the other hand, for logarithmic functions we need to equate the argument of the function to 1 and solve for x . Finally, we substitute the value of x into the function and find the value of $f(x)$.

Problem 4: Example of exponential functions

Lets find the critical point of

$$f(x) = \exp[e \ln(x) - 2e]$$

Since the argument is $e \ln(x) - 2e$, let's equate to 0 and solve for x .

$$\begin{aligned}
 e \ln(x) - 2e &= 0 \\
 e \ln(x) - 2e + 2e &= 0 + 2e \\
 e \ln(x) &= 2e \\
 \frac{1}{e} e \ln(x) &= \frac{1}{e} 2e \\
 \ln(x) &= 2 \\
 e^{\ln(x)} &= e^2 \\
 x &= e^2
 \end{aligned}$$

Now that we know the value of x , we can substitute this value into the original equation to find the value of $f(e^2)$.

$$\begin{aligned}
 f(x) &= \exp[e \ln(x) - 2e] \\
 f(e^2) &= \exp[e \ln(e^2) - 2e] \\
 &= \exp[2e \ln(e) - 2e] \\
 &= \exp[2e - 2e] \\
 &= \exp[0] = e^0 \\
 &= 1.
 \end{aligned}$$

Hence the critical point is at $(e^2, 1)$.

Critical point at : $(e^2, 1)$

Problem 5: Example of logarithmic functions

Let's find the critical point of

$$g(x) = \ln[\exp[x^2] - 8].$$

Since the argument is $\exp[x^2] - 8 = e^{x^2}$, lets equate to 1 and solve for x .

$$\begin{aligned}
 e^{x^2} - 8 &= 1 \\
 e^{x^2} - 8 + 8 &= 1 + 8 \\
 e^{x^2} &= 9 \\
 \ln[e^{x^2}] &= \ln[9] \\
 x^2 \ln[e] &= \ln[9] \\
 x^2 &= \ln[9] \\
 (x^2)^{1/2} &= (\ln[9])^{1/2} \\
 x^{\frac{2}{2}} &= (\ln[9])^{1/2} \\
 x &= \sqrt{\ln[9]}.
 \end{aligned}$$

Now that we know the value of x , we can substitute this value into the original equation to find the value of $f(\sqrt{\ln[9]})$.

$$\begin{aligned}
 g(x) &= \ln[\exp[x^2] - 8] \\
 f(\sqrt{\ln[9]}) &= \ln\left[\exp\left[\left(\sqrt{\ln[9]}\right)^2\right] - 8\right] \\
 &= \ln\left[\exp\left[\left(\ln[9]\right)^{1/2}\right]^2 - 8\right] \\
 &= \ln\left[\exp\left[\left(\ln[9]\right)^{2/2}\right] - 8\right] \\
 &= \ln[\exp[\ln[9]] - 8] \\
 &= \ln[9 - 8] \\
 &= \ln[1] \\
 &= 0
 \end{aligned}$$

Hence the critical point is at $(\sqrt{\ln[9]}, 0)$.

Critical point at : $(\sqrt{\ln[9]}, 0)$

4 Modelling with exponential and logarithmic functions

In this section are applications of the exponential and logarithmic functions.

Problem 6: Example of logarithmic functions functions

The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 22mg. Recalling that the mass decays exponentially, fulfill the following points

- Write the function that models the remaining mass after t years
- How much mass will remain after 4000 years?
- How much time will it take for the sample to be only 10 mg.

Writing the function that models the remaining mass after t years Let's recall the algebraic expression for exponential decay,

$$m(t) = m_o e^{-rt}.$$

Where $m(t)$ is the mass at a time t , m_o is the mass at the time $t = 0$, r is the rate of decay and t is time. We need to find the numeric values of m_o and r in order to write the function that models the remaining mass after t years. To find r we can use the sentence *The half-life of Radium-266 is 1600 years*, to write the following equation,

$$m(1600) = \frac{m_o}{2} = m_o e^{-r \cdot 1600}.$$

Although the initial mass was given, we don't need it, because m_o cancels out during the process of solving for r ,

$$\begin{aligned} \frac{m_o}{2} &= m_o e^{-rt} \\ \frac{1}{m_o} \frac{m_o}{2} &= \frac{1}{m_o} m_o e^{-rt} \\ \frac{1}{2} &= e^{-r \cdot 1600} \\ \ln \left[\frac{1}{2} \right] &= \ln \left[e^{-r \cdot 1600} \right] \\ \ln[1] - \ln[2] &= (-r \cdot 1600) \ln[e] \\ \cancel{\ln[1]}^0 - \ln[2] &= (-r \cdot 1600) \cancel{\ln[e]}^1 \\ -\ln[2] &= (-r \cdot 1600) \\ -\frac{1}{1600} \ln[2] &= -r \cdot 1600 \cdot \frac{1}{1600} \\ -\frac{1}{1600} \ln[2] &= -r \\ -\frac{1}{1600} \ln[2] \cdot -1 &= -r \cdot -1 \\ \frac{1}{1600} \ln[2] &= r \end{aligned}$$

By finding the numeric value using a calculator we get $r \approx 4.332\,169 \times 10^{-4}$. Now that we know the rate, we only need to identify the initial mass from the given information. From the sentence *Suppose that there is a sample of 22mg*, we can say that $m_o = 22$. Therefore the function that models the remaining mass after t years is,

$$m(t) = 22e^{-4.332\,169 \times 10^{-4}t}.$$

A more elegant expression will be,

$$\begin{aligned} m(t) &= 22e^{-\frac{1}{1600} \ln[2]t} \\ &= 22 \left(e^{\ln[2]} \right)^{-\frac{1}{1600}t} \\ &= 22(2)^{-t/1600} \end{aligned}$$

How much mass will remain after 4000 years? We are going to use both equations to find the remaining mass at time $t = 4 \times 10^3$ years. Let's start with the conventional one,

$$\begin{aligned} m(4 \times 10^3) &= 22e^{-4.332\,169 \times 10^{-4} \cdot 4 \times 10^3} \\ &= 22e^{-(4.332\,169 \cdot 4) \times 10^{-4+3}} \\ &= 22e^{-(17.328676) \times 10^{-1}} \\ &= 22e^{-1.7328676} \\ &= 22e^{-1.7328676} \\ &\approx 3.889088 \end{aligned}$$

$$m(4 \times 10^3) \approx 3.889088.$$

Now let's use the elegant result,

$$\begin{aligned} m(4 \times 10^3) &= 22(2)^{-4 \times 10^3 / 1600} \\ &= 22(2)^{-2.5} \\ &\approx 3.889087 \end{aligned}$$

We can see that are almost the same, where the difference is 0.000001.

How much time will it take for the sample to be only 10 mg finally, we are going to solve for t in order to find the time at which $m(t) = 10$. We start with the conventional answer,

$$\begin{aligned}
 10 &= 22e^{-4.332169 \times 10^{-4}t} \\
 10 \frac{1}{22} &= 22 \frac{1}{22} e^{-4.332169 \times 10^{-4}t} \\
 \frac{10}{22} &= e^{-4.332169 \times 10^{-4}t} \\
 \ln \left[\frac{5}{11} \right] &= \ln \left[e^{-4.332169 \times 10^{-4}t} \right] \\
 \ln \left[\frac{5}{11} \right] &= (-4.332169 \times 10^{-4}t) \ln[e] \\
 \ln \left[\frac{5}{11} \right] &= -4.332169 \times 10^{-4}t \\
 \frac{1}{-4.332169 \times 10^{-4}} \ln \left[\frac{5}{11} \right] &= \frac{1}{-4.332169 \times 10^{-4}} \cdot -4.332169 \times 10^{-4}t \\
 \frac{1}{-4.332169 \times 10^{-4}} \ln \left[\frac{5}{11} \right] &= t \\
 1820.207264 &\approx t
 \end{aligned}$$

$$m(1820.207264) \approx 10.$$

Finally, let's use the elegant expression,

$$\begin{aligned}
 10 &= 22 (2)^{-t/1600} \\
 \frac{1}{22} 10 &= \frac{1}{22} 22 (2)^{-t/1600} \\
 \frac{10}{22} &= 2^{-t/1600} \\
 \log \left[\frac{5}{11} \right] &= \log \left[2^{-t/1600} \right] \\
 \log \left[\frac{5}{11} \right] &= \left(-\frac{t}{1600} \right) \log [2] \\
 \log \left[\frac{5}{11} \right] \frac{1}{\log [2]} &= \left(-\frac{t}{1600} \right) \log [2] \frac{1}{\log [2]} \\
 \frac{\log \left[\frac{5}{11} \right]}{\log [2]} &= \left(-\frac{t}{1600} \right) \\
 -1600 \cdot \frac{\log \left[\frac{5}{11} \right]}{\log [2]} &= (-1600) \left(-\frac{t}{1600} \right) \\
 -1600 \cdot \frac{\log \left[\frac{5}{11} \right]}{\log [2]} &= t \\
 1820.005638 &\approx t
 \end{aligned}$$

We see now that the difference is 0.201626 years, that would be 78.9349 days or 78 days, 22 hours, 26 minutes and 15 seconds.